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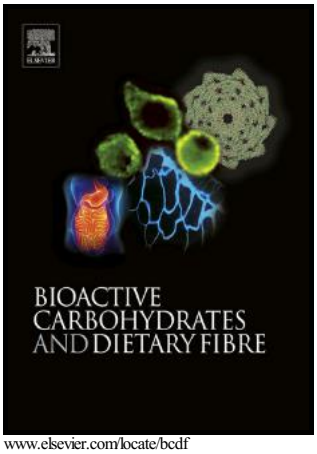

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Title: Mushroom polysaccharides as potential prebiotics with their antitumor and immunomodulating properties: A review

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ABSTRACT

Mushrooms seem to be a potential source for prebiotics as they contain different polysaccharides such as chitin, hemicellulose, α- and β-glucans, mannans, xylans and galactans. Among mushroom polysaccharides, β-(1→3)-D-glucans, and their peptide/protein derivates (polysaccharide-peptide/protein complexes), proteoglycans are essential prebiotics and found to play vital role in immunomodulating and antitumor activities. These prebiotic compounds display immunomodulating and antitumor activity similar to those effected by immune effector cells such as lymphocytes, macrophages, hematopoietic stem cells, T cells, dendritic cells (DCs) and natural killer (NK) cells that play essential roles in innate and adaptive immunity, resulting in production of biologic response modifiers. Several glucans and heteroglycans were found with their significant immunoenhancing properties which could stimulate the macrophages, splenocytes and thymocytes. Thus this review aims to summarize and explore the potential of mushroom polysaccharides as prebiotics with their antitumor and immunomodulating properties for the development of nutraceutical foods and drugs.

Keywords: Mushroom; Polysaccharides; Prebiotics; Immunomodulation; Antitumor

1. Introduction

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Prebiotics like oligosaccharides and polysaccharides have garnered much attention as functional foods advancing the growth of beneficial lactic acid bacteria in the colon while exercising antagonistic action on harmful bacteria like *Salmonella* sp. or *Escherichia coli*, thus, discouraging their proliferation to improve host health (Gibson, 2004). Dietary carbohydrates such as fibers have been previously known to be utilized as prebiotics (Blaut, 2002), but recently, the concept of oligosaccharides has given a whole new dimension to the development of prebiotics (Gibson, 2004). Currently available prebiotics such as inulin and its derivatives as well as galacto-oligosaccharides (GOS) are relatively cheaper to manufacture and have been widely used as functional ingredients in food (Macfarlane et al., 2006). Chicory root derived-inulin (FOS), wheat bran-derived arabinoxylooligosaccharides (AXOS) and xylooligosaccharides (XOS) have been proved to have huge applications in human health (Sabater-Molina et al., 2009; Femia et al., 2010).

In recent times, food is intended not just satiate hunger and supply essential nutrients, but also to bring about improvements in health. With respect to this, functional foods have a significant role to play (Siro et al., 2008). According to the Institute of Medicine's Food and Nutrition Board, "Functional Foods" are foods or dietary components that may provide a health benefit beyond basic nutrition. Prebiotics such as oligosaccharides and polysaccharides (inulin) have garnered great interest as functional food ingredient because these are able to manipulate the composition of colonic microbiota in human gut by inhibition of exogenous pathogens (Rycroft et al., 2001), thus improving host health (Roberfroid, 2000; 2002) Extensive research on prebiotics has elevated the need to look for alternative sources of prebiotics. Mushroom seems to be a potential candidate for prebiotics as it contains carbohydrates like chitin, hemicellulose, β and α-glucans, mannans, xylans and galactans. Chitin, a water insoluble polysaccharide is indigestible in human system and thus plays the role of a dietary fiber (Cheung, 1996; Kalac, 2009). However the bioactivities of water insoluble polysaccharides have been revealed to be less than that of water soluble polysaccharides (Tao et al., 2006). Most polysaccharides from mushroom sources exist as linear and branched glucans with varying types of glycosidic linkages such as (1→3), (1→6)-β-glucans and (1→3)-α-glucans. However, some of them are true heteroglycans containing arabinose, mannose, fructose, galactose, xylose, glucose and glucuronic acids as main side chain components or in different combinations (Wasser, 2002; Giavasis & Biliaderis, 2006). Despite being of different chemical compositions, most mushrooms polysaccharides belong to the group of β-glucans (Wasser, 2002). Digestive enzymes secreted by the pancreas (e.g. trypsinogen, amylase) or brush border of vertebrates (e.g. lactase, peptidase) and of mammals in particular, are unable to hydrolyze β-glucosidic bonds of β-glucans. This makes them resistant to acid hydrolysis in the stomach and thus these compounds remain non-digestible by human digestive enzymes (Van Loo, 2006). The unique property of mushroom carbohydrate to be non-digestible, increases its likelihood to be a prospective prebiotic, keeping in mind the definition of a prebiotic. Nevertheless, exhaustive studies need to be carried out to establish such a claim, because not all dietary carbohydrates are prebiotics (Gibson et al., 2004).

Mushroom polysaccharides are not only important as prebiotics but have immense other biological properties such as antitumor, antimicrobial, antioxidant, antiviral, and immunomodulating activities (Lindequist et al., 2005; Thatoi & Singdevsachan, 2014). Many polysaccharides have been isolated from mushrooms and extensively studied for their antitumor properties. Three important polysaccharides; ''Krestin" (PSK) from the cultured mycelium of Kawaratake

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(*Trametes versicolor*), ‘‘Lentinan” from the fruiting bodies of Shiitake (*Lentinus edodes*) and ‘‘Schizophyllan” (Sonifilan) from the culture fluid of Suehirotake (*Schizophyllum commune*) have been isolated and developed as cytostatic polysaccharide drugs in Japan (Mizuno, 1993). Lentinan and Schizophyllan are pure β-glucans, whereas PSK is a protein bound polysaccharide (Larone, 2002). The biological activity of these three products is related to their antitumor and immunomodulating properties, which enhance the body’s defenses against various forms of infectious disease. Along these lines the present review aims to summarize and explore the potential of mushroom polysaccharides as prebiotics with their antitumor and immunomodulating properties for the development of nutraceutical foods and drugs.

2. Concept of prebiotic and their mechanism action

The term prebiotic was actually given by Gibson and Roberfroid (1995) and defined as ‘‘a non-digestible food ingredient that beneficially affects the host by selectively stimulating the growth and/or activity of one or a limited number of bacteria in the colon”. The concept of prebiotics is laid out by certain criteria as mentioned by Gibson et al. (2004) viz. resistance to gastric acidity, hydrolysis by mammalian enzymes and gastrointestinal absorption; fermentation by intestinal microflora and selective stimulation of the growth, and/or activity of intestinal bacteria associated with host well-being and health. Further, Wang (2009) has highlighted and pointed out the important criteria of a food ingredient as prebiotic (Fig. 1). The first attribute of prebiotics, which is non-digestible or resistant to upper gut tract indicates that the prebiotics can withstand digestive processes before they reach the colon, thus stimulating beneficial bacteria like bifidobacteria and lactobacilli effectively (Gibson & Collins, 1999; Macfarlane et al., 2008). Criteria which allow the classification of a food ingredient as a prebiotic also include selective fermentation by potentially beneficial bacteria in the colon (Gibson et al., 2004; Wang, 2009). The effects of this fermentation may lead to an increase in the expression or change in the composition of short-chain fatty acids, increased fecal weight, a mild reduction in luminal colon pH, a decrease in nitrogenous end products and reductive enzymes, an increased expression of the binding proteins or active carriers associated with mineral absorption and immune system modulation (Douglas & Sanders, 2008; Śliz□ewska et al., 2012) (Fig. 2), which is beneficial to the host health; requirement for the third criteria. Selective stimulation of the growth and/or activity of intestinal bacteria potentially associated with health and well-being is considered as one of the quintessential criteria of prebiotics (Gibson et al., 2004). Probiotics are live microorganisms that when administered in adequate amounts confer a health benefit on the host (FAO/WHO report, 2001). Prebiotics are markedly suitable for the growth and activities of probiotics, bifidobacteria and lactobacilli (Wang, 2009) while suppressing the growth of clostridia and bacteroides. Yet, this is the toughest parameter to be met because not all candidate prebiotics show selective fermentation. Studies conducted by Langlands et al. (2004) and Duncan et al. (2003) have brought out that fermentation of the prebiotic-inulin has caused an increase in other gut bacterial genera namely, *Roseburia*, *Ruminococcus* and *Eubacterium*. This is because different people harbor different bacterial species and the composition of the microbiota can be affected by a variety of other factors such as diet, disease, drugs, antibiotic, age and others (Macfarlane et al., 2006). Last but not the least, a prebiotic must be able to withstand food processing conditions so that they remain intact without being degraded or chemically altered while being available for bacterial

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metabolism in gut (Huebner et al., 2008). As a result, the gastrointestinal health of human beings can be improved (Tuohy et al., 2003).

Gibson (2004) found dietary carbohydrates like fibers to be important prebiotics, but oligosaccharides were observed to be rather more promising. Stowell (2007) extensively reviewed the existing prebiotics and categorized them basing on certain common criteria. Inulin, fructooligosaccharides, galactooligosaccharides, lactulose and polydextrose are sorted to be "established prebiotics", whereas isomaltooligosaccharides, xylooligosaccahrides and lactitol are grouped under "emerging prebiotics". Other than the aforementioned prebiotic compounds, mannitol, maltodextrin, raffinose, and sorbitol are also prebiotics with proven health properties (Mandal et al., 2009; Vamanu & Vamanu, 2010). Starch-rich whole grains are considered to have natural prebiotic properties and as a conjecture, their consumption may lead to several health benefits. Prebiotic consumption is thought to enhance immune function, improve colonic integrity, decrease incidence and duration of intestinal infections, decrease the amount of long chain fatty acids in bowels, decrease pH values in bowels, raise the production of short chain fatty acids in down-regulated allergic responses as well as boost digestion and elimination of faeces (Douglas & Sanders, 2008; Śliz□ewska et al., 2012) (Fig. 2). However, these effects are not the immediate consequences of prebiotic ingestion. Wang (2009) suggested that the effect of prebiotics is in fact indirect, because it is the changes occurring in the gastrointestinal microbiota compositions (*Bifidobacteria*, *Lactobacilli*, as well as the *Histolyticum* subgroup; bacteroides and clostridia) that in reality gives rise to the positive impact of prebiotics. *Bifidobacteria* and *Lactobacilli* are the beneficial bacteria that serve as prebiotics target (Macfarlane et al., 2008). One important positive effect imparted by prebiotics is reflected in the significant increase in numbers of probiotics such as *Bifidobacteria* and *Lactobacilli*, while retarding the development of *Histolyticum* subgroup (Palframan et al., 2003). Reports by Probert & Gibson (2002), Langlands et al. (2004) as well as Macfarlane et al. (2006) bring out the increase in number of *Bifidobacteria* and *Lactobacilli* in the gut, as a result of prebiotic action. Gibson et al. (1995) found that *Bifidobacteria* was able to stimulate the immune system, produce vitamin B, inhibit pathogen growth, reduce blood ammonia and blood cholesterol levels and also restore the normal flora after antibiotic therapy, while the *Lactobacilli* aided in digestion of lactose in lactose-intolerant individuals, reducing constipation and infantile diarrhea, increasing resistance to infections caused by *Salmonellae* and helping to relieve irritable bowel syndrome (Manning & Gibson, 2004). Potential and known mechanisms of probiotics have been described by O'Toole and Cooney (2008) for their beneficial effects which include (i) impact of prebiotics upon the composition of the microbiota apparently by competing with them for substrate availability, and (ii) alteration of the dynamics of carbohydrate utilization by individual microbiota components; (iii) production of vitamins and their enhanced availability which may modulate the microbiota; (iv) effect of probiotics bacteria on the general microbiota probably by direct antagonism; (v) increase in competitive exclusion for binding sites of pathogens; (vi) improvement in barrier function of epithelial cell; (vii) reduction of inflammation, thus altering intestinal properties for colonization and persistence within; (viii) stimulation of innate immune response (Fig. 3).

3. Mushrooms as an important source of polysaccharide

Recently, polysaccharides from various foods, such as cereal (Michida & Pizzoferrato, 2006), oats (Gokavi et al., 2005), mushrooms (Guo et al., 2004; Aida et

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al., 2009), herbs (Guo et al., 2004), chicory root (Roberfroid, 2000), citrus (Sendra et al., 2008), soybeans (Crittendan & Playne, 1996) and potatoes (MacFarlane et al., 2006), are gaining attention as new alternatives for development of food supplements to increase the health benefits. Among these, edible mushrooms are noteworthy of being foods used as nutritional supplements, and their safety has also been well established. Mushrooms are highly nutritive with high content of carbohydrate, proteins, vitamins, minerals, fibers and low/no calories and cholesterol (Thatoi and Singdevsachan, 2014). There are a couple of research studies in support of the fact that polysaccharides from *Pleurotus* spp. (Synytsya et al., 2009), *Lentinus edodes*, *Tremella fuciformis* (Guo et al., 2004) and *Agaricus bisporus* mushroom (Giannenas et al., 2011) have been successfully used as nutraceuticals. Amongst these edible mushrooms, *Pleurotus eryngii*, *L. edodes* and *Flammulina velutipes* contain ribose, xylose, fructose, mannose, glucose and trehalose. Sucrose was observed only in *L. edode* whereas glucose was primarily present in both *P. eryngii* and *L. edode*. *F. velutipes* contains mainly xylose (Kim et al., 2009).

Different mushrooms produce different types of polysaccharides that can be either water soluble or water insoluble. Some polysaccharides have only glucose moiety while some have proteins attached with them. Polysaccharides are mostly β-linked glucose molecules but some also have galactose and mannose; yet some are heteropolysaccharides and others are glucan-protein complexes. These polysaccharides can be utilized in treatment of simple to complex diseases like cancer, AIDS and other present day diseases (Ooi and Liu, 2000).

4. Mushroom polysaccharides as prebiotics

The balance of intestinal flora is crucial for human health and disease prevention. In this context, prebiotics are the most promising health foods because they can regulate the structure and number of intestinal flora. Mushrooms (*Pleurotus ostreatus* and *Lentinus edodes*) can significantly modify intestinal flora composition by promoting the metabolism and proliferation of beneficial microorganisms such as *Lactobacilli* and *Bifidobacteria*, as well as by inhibiting pathogenic bacteria such as *E coli*, *Clostridium* and *Salmonella* (Zhou et al., 2011). The major components rendering prebiotic function in mushroom are non-digestible polysaccharides such as glucan, chitin and hetropolysaccharides. Mushrooms also prevent viral infection by enhancing the growth of probiotic bacteria in the large intestine (Villares et al., 2012). Several mushroom polysaccharides like pleuran, lentinan, schizophyllan, β and α-glucans, mannans, xylans, galactans, chitin, inulin and hemicelluloses can be credited to promising prebiotic effects (Table 1). Pleuran from oyster (*Pleurotus ostreatus*) mushrooms and lentinan from Shiitake (*Lentinus edodes*) mushrooms are currently the most frequently used β- glucans as prebiotics. Both of them show positive effects on the intestines. They increase the resistance of intestinal mucosa to inflammation (Zeman et al., 2001) and inhibit the development of intestinal ulcers in rats (Nosalova et al., 2001). Lentinan also shows a positive effect on peristalsis in weaned piglets (Van Nevel et al., 2003). Besides, it has been demonstrated that sclerotial β-glucans from mushrooms (*Pleurotus tuber- regium*, *Polyporous rhinocerus* and *Wolfiporia cocos*) can be utilized by human colonic bacteria *in vitro* and therefore they have the potential to be used as novel prebiotics that can influence human gut health by selectively modulating the growth of probiotic bacteria including bifidobacteria and lactic acid bacteria (Wong et al., 2005). The colonic fermentation of sclerotial β-glucans isolated from *P. tuber-regium* could also enhance the absorption of calcium and magnesium in ovariectomized rats which might have the health implication of

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improving mineral absorption in the human gut (Wong et al., 2006). Synytsya et al. (2009) gave a positive overview that mushroom glucans of *Pleurotus ostreatus* and *P. eryngii* were able to stimulate the growth of probiotics - *Lactobacillus* sp. (4 strains: Lac A-D), *Bifidobacterium* sp. (3 strains: Bifi A-C) and *Enterococcus faecium* (2 strains: Ent A and B) - to some extent. Maximum growth rate, maximum biomass concentration and final acid production were observed in the study. It was found that glucan from *P. eryngi* supported the growth of *Lactobacillus* strains better than *P. ostreatus*. *Lactobacillus* B and C showed the highest production of short chain fatty acid (SCFA), while *Bifidobacteria* A showed the lowest amount of SCFA when supplemented with both extracts. However, glucan of *Pleurotus ostreatus* and *P. eryngii* can be used as prebiotic compound for human gut health maintenance.

Additionally, synergetic effects of mushroom polysaccharides have been demonstrated by several researchers for the growth of probiotics. Effect of three kinds of crude polysaccharides (PS I, PS II, PS III) from *Agaricus blazei* Murill (obtained by diethyl- aminoethanol cellulose column chromatography) on the growth of lactic acid bacteria have been studied by Lili and Jianchun (2008). It was found that PS I and PS II had better growth promoting effect on lactic acid bacteria than PS III and they all promoted lactic acid production by *Lactobacillus acidophilus*. Further, effect of crude polysaccharides extracted from *Ganoderma lucidum* on the growth of probiotics have been studied by Yamin et al. (2012) in batch-culture fermentation of human faecal culture. Growth promotion of *Bifidobacterium* sp. and *Lactobacillus* sp. and growth inhibition of *Salmonella* sp. prove its prebiotic effect. These results certainly increase the possibility for a new prebiotic source.

Recently, Chou et al. (2013) studied the prebiotic activity of crude polysaccharides from *Lentinula edodes* stipe, *Pleurotus eryngii* base, and *Flammulina velutipes* base which were found to enhance the survival rate of probiotics (*Lactobacillus acidophilus*, *Lactobacillus casei*, and *Bifidobacterium longum* subsp. *longum*) during cold storage. The polysaccharides displayed synergistic effects with the peptides and amino acids in a yogurt culture to maintain probiotics above 107 CFU/mL during cold storage. They also had significant protective effects on these probiotics in simulated gastric and bile juice conditions to achieve beneficial effects in the host.

5. Mushroom polysaccharides with their antitumor and immunomodulatory activities

5.1. Glucan

The polysaccharides of mushrooms occur mostly as glucans, some of which are linked by β-(1-3), (1-6) glycosidic bonds and α-(1-3) glycosidic bonds but most of these polysaccharides are true heteroglycans. Hetero-β-D-glucan is a repeating structure, with its D-glucose molecules joined together in linear chains by beta-bonds (β). These can extend from the carbon 1 of one saccharide ring to the carbon 3 of the next (β1-3), from carbon 1 to carbon 4 (β1-4), or from carbon 1 to carbon 6 (β1-6). Most often there is a main chain which is either β1-3, β1-4, or mixed β1-3, β1-4 with β1-6 side chains. Glucans are important for their marked therapeutic properties. From the fruiting bodies of *Agaricus blazei,* a water-soluble β-(1→6)-D-glucan protein complex was isolated in addition to water-soluble β-(1→3)-glucan. The first marked anticancer activity was noted on β-(1→6)-D-glucan (Mizuno, 1995). The β-1,3;1,6-glucans from higher fungi and yeast became a new biological entity, the so-called "biologic response modifiers" that function as immunostimulants against infectious diseases and show a possible tumoricidal activity (Firenzuoli et al., 2008).

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Composition-wise, antitumor polysaccharides majorly resemble fungal cell walls in that; they consist of polysaccharides such as cellulose, (1→3), (1→6)-β-glucans and (1→3)-α-glucans or polysaccharide-protein complexes such as galactomannan-protein and glucuromannan-protein (Stone & Clarke, 1992). Highly active polysaccharides from mycelia are predominantly protein-containing glucans having molecular weight of 10,000 kDa (Mizuno et al., 1995; Zhang et al., 1994). It has been stated that the β-(1-3) linkages in the backbone (main chain) of the glucan and additional β-(1-6)-branch points are required for antitumor activity (Wasser, 2002). Ganoderan, an immunomodulatory β-glucan produced by *G. lucidum* brings about appreciable antitumor immunity in tumour-bearing mice (Ooi & Liu, 2000).

An α-1,6 and α-1,4 glucan complex and a glucomannan with a main chain of β-1,2-linked D-mannopyranosyl residues have been isolated from *Agaricus blazei* Murril and found to inhibit tumorigenesis (Mizuno et al., 1998). Responses to the varied polysaccharides are possibly accredited to the different cell-surface receptors, which may be present only on particular subsets of cells, and may evoke distinct downstream responses. A combination of such responses involving different cell subsets could conceivably provide greater tumor inhibition than could be induced by a single polysaccharide (Firenzuoli et al., 2008). Polysaccharides having immunomodulating capacity, only those which consist of a (1→3)-linked β-glucan backbone with (l→6)-linked β-D-glucopyranosyl units as branches produce complete inhibition of tumor growth. (1→3)-β-glucans from fungi commonly have a tumor inhibition percentage of 99-l00%, while other polysaccharides exhibit l0-40% inhibition (Blaschek et al., 1992). There are also reports that some mushroom β-1,3; 1,6-glucans could mediate tumor regression when given orally (Suzuki et al., 1991 and Cheung et al., 2002). In more recent studies using human tumor xenografts, orally administered soluble barley β-1,3;1,4-glucan or *in vitro* antitumor monoclonal antibodies were ineffective as single agents, but when combined, elicited a substantial antitumor effect (Cheung and Modak, 2002). β-glucans can potentially be used to generate a novel cell-mediated effector mechanism for tumor vaccines and antibodies to tumor antigens that otherwise rely mostly on the direct cytotoxic action of chemotherapy.

A β-D-glucan polysaccharide isolated from *Agaricus brasiliensis* exhibited immunostimulative and antitumour activity (Mizuno et al., 1995). However a higher antitumour activity was observed in another xyloglucan protein complex obtained from 5% NaOH solution (Mizuno et al., 1995). A glycoprotein fraction obtained from *A. campestris* also exhibited antitumour activity against sarcoma 180 in ICR mice; the protein moiety was composed of 17 amino acids (Jeong et al., 1990). Along with ergosterol, six other steroids were also isolated from an acetone extract of *A. brasiliensis* fruit bodies; out of these, three of them effectively inhibited cell proliferation of cervical cancer cell (HeLa cells) (Mizuno, 2002). The antitumour glycoprotein proflamin found in mycelia of *F. velutipes* is effective against allogenic and syngenic tumours by oral administration. Proflamin augments antibody formation and activates lymphocyte blastogenesis (Ikekawa et al., 1985 and Ikekawa, 1995). A protein with β-glucan isolated from *Grifola frondosa* extract exhibits antitumour activity by potentiating anti-tumour cellular functions by directly enhancing various mediators such as lymphokines and IL-1 (Nanba, 1993). Several studies have shown that β-D-glucan derived from *Grifola frondosa* have strong antitumor activity in xenographs (Kurashiga et al., 1997).

Unlike most other natural products, purified β-1,3-glucans retain their bioactivity. β-1,3-glucans can work at a cellular and molecular level, functioning

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through stimulation of granulocytes (neutrophils and eosinophils), monocytes, macrophages and NK-cells (Brown and Gordon, 2001). The biochemical importance of β-(1→3)-side branches in glucan has been confirmed and has been shown to be associated with the enhancement of the immunomodulatory activity of polysaccharides (Dong et al., 2002). Certain data also suggested that β-glucans could promote T cell-specific responses, perhaps, through triggering the secretion of IFN-ɣ , IL-6, IL-8 and IL-12 from macrophages, neutrophils and NK-cells (Firenzuoli et al., 2008). Glucan from edible mushrooms have been found to have immunomodulatory activity, encouraging stimulation of macrophages, splenocytes and thymocytes. An immunoenhancing water soluble glucan (Fig. 4) of an edible mushroom *Pleurotus florida*, cultivar Assam Florida, has been observed to be constituted of only D-glucose as a monosaccharide constituent which is able to stimulate macrophages, splenocytes and thymocytes (Roy et al., 2009). Glucan (Fig. 5) from *Lentinus squarrosulus* (Mont.) Singer consists of (1→3,6)-linked, (1→3)-linked, (1→6)-linked and terminal β-D-glucopyranosyl moieties in a relative proportion of approximately 1:2:1:1 and showed optimum activation of macrophages as well as splenocytes and thymocytes (Bhunia et al., 2011). Another immunostimulating glucan, an (1→4)-, (1→6)-branched glucan (Fig. 6) from *Calocybe indica* also invigorates the splenocytes and thymocytes (Mandal et al., 2012). Other immunostimulating glucans from mushrooms are PS-I (1→6-β-D-glucan) and PS-II (1→3,6-β-D-glucan) from *Termitomyces robustus* var. (Bhanja et al., 2012), glucan (Fig. 7) from an ectomycorrhizal edible mushroom *Russula albonigra* (Krombh.) Fr. (Nandi et al. 2012) and glucan obtained from *Ramaria botrytis* (Fig. 8) (Bhanja et al., 2013). Glucan (Fig. 9) isolated from *Tricholoma crassum* (Berk.) Sacc. is also noteworthy in that it showed macrophage activation *in vitro* by NO production in a dose dependent manner and strong splenocyte and thymocyte immunostimulation in mouse cell culture medium (Samanta et al., 2013).

Further, glucan from hybrid mushrooms also exhibit immunomodulatory activity. A (1→6)-β-D-glucan from a somatic hybrid mushroom (*PfloVv5FB*) of *Pleurotus florida* and *Volvariella volvacea* exhibit significant immunoenhancing properties which could stimulate the macrophages, splenocytes and thymocytes (Das et al., 2010). Another (1→6)-β-D-glucan from a hybrid mushroom (obtained by backcross mating between hybrid mushroom *PfloVv12* and *Volvariella volvacea*) has been also found to be immunostimulating in nature (Sarkar et al., 2012). Other similar examples include a water soluble glucan (comprising terminal (1→3,6)-linked and (1→6)-linked β-D-glucopyranosyl moieties in a molar ratio of nearly 1:1:3) isolated from an edible hybrid mushroom (*Pfle1r*) of *Pleurotus florida* and *Lentinula edode* (Maji et al., 2012) and a (1→3,6)-β-D-glucan from hybrid mushroom (*PfloVv5FB*) (obtained through protoplast fusion between *Pleurotus florida* and *Volvariella volvacea* strains) (Maity et al., 2013).

5.2. Heteropolysaccharides and glycoproteins

Hetero-β-D-glucans, i.e., linear polymers of glucose with other D-monosaccharides, can have anticancer activity (Wasser, 2002). Mushrooms also contain β-D-glucans with heterosaccharide chains of xylose, mannose, galactose and uronic acid extracted by salt and alkali and β-D-glucan-protein complexes that are present at 10% to 50% in dry matter. Heteroglucan side chains contain glucuronic acid, galactose, mannose, arabinose or xylose either individually as a main component or in different combinations. Heteropolysaccharides have molecular weight of about 10,000 kDa and consist of galactose, glucose, mannose and fucose (Zhang et al., 1994; Mizuno et al.,

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1995). Some of the heteropolysaccharides have been shown to be carcinostatic when given by intraperitoneal injection or by oral administration. Antitumor polysaccharides isolated from fruiting bodies and having antiproliferative activity are mostly heteropolysaccharides (Zhang et al., 1994; Mizuno et al., 1995). Heteroglycan-protein complexes from *Grifola frondosa* are found to inhibit tumour growth by stimulating the immune function as a biological response modifier (Nanba et al., 1987 and Cun et al., 1994).

Several heteroglycans have been reported to impart potential immunomodulatory effects. A heteroglycan (Fig. 10) isolated from the hot water extract of the fruiting bodies of an edible mushroom, *Lentinus squarrosulus* (Mont.) Singer contains D-galactose, L-fucose, and D-glucose in a molar ratio of nearly 1:1:5 and this polysaccharide is capable of activating macrophages, splenocytes and thymocytes (Bhunia et al., 2010). Heteroglycan (Fig. 11) from an edible mushroom *Calocybe indica* var. APK2 comprises D-glucose, D-galactose, and L-fucose in a molar ratio of nearly 3:1:1 and has been proven to show immunoenhancing and cytotoxic activity upon HeLa cell lines (Mandal et al., 2011). Heteroglycans (Fig.12) extracted from the hot aqueous extract of *Pleurotus ostreatus* cultivar containing D-glucose and D-galactose in a molar ratio of nearly 7:1 is also an immunostimulant (Maity et al., 2011). Heteroglycans from hybrid mushrooms also exhibit immunoenhancing activity. Some examples of these are (a) heteroglycan (Fig. 13) from an aqueous extract of the somatic hybrid mushroom (*PfloVv1aFB*) of *Pleurotus florida* and *Volvariella volvacea,* constituted of D-galactose, D-mannose, and D-glucose with the molar ratio of nearly 1:1:4 (Patra et al., 2011) (b) heteroglycan (Fig. 14) from an alkaline extract of a somatic hybrid mushroom (*PfloVv1aFB*) of *Pleurotus florida* and *Volvariella volvacea* and consisting of D-galactose, D-mannose, and D-glucose with the molar ratio of nearly 1:1:4 (Bhunia et al., 2012) (c) heteroglycan (Fig. 15) isolated from hot aqueous extract of fruiting bodies of an edible hybrid mushroom *Pfle1r* of *Pleurotus florida* and *Lentinula edodes* and made up of D-glucose, D-mannose, and D-galactose residues in a molar ratio of nearly 1:1:1 (Maji et al., 2013) (d) an immunostimulating water-soluble heteroglycan (PS-II) (Fig. 16) isolated from aqueous extract of the fruiting bodies of a hybrid mushroom, *Pfls1h* of *Pleurotus florida* and *Lentinus squarrosulus* (Mont.) Singer. composed of (1→6)- and (1→2,4,6)-α-D-galactopyranosyl, terminal β-D-mannopyranosyl and terminal β-D-glucopyranosyl residues in a relative proportion of approximately 1:1:1:1 (Sen et al., 2013). An immunoenhancing water-soluble hetero-polysaccharide (Fig. 17) isolated from an alkaline extract of the fruit bodies of an ectomycorrhizal edible mushroom, *Tricholoma crassum* (Berk.) Sacc. exhibited splenocyte, thymocyte as well as macrophage activations (Patra et al., 2012). Heteropolysaccharide (Fig. 18) having molecular weight ~2.1×10^5 Da was isolated from hot aqueous extract of the fruit bodies of hybrid mushroom *Pfle 1p*. The hybrid mushroom *Pfle 1p* was obtained through intergenic protoplast fusion between *Pleurotus florida* and *Lentinula edodes*. The heteropolysaccharide contained D-glucose, D-galactose, and D-mannose in a molar ratio of nearly 4:2:1 and showed *in vitro* macrophage activation by NO production and also stimulated splenocytes and thymocytes (Maity et al., 2013).

5.3. Lentinan

Lentinan, produced from the fruiting bodies of Shiitake mushroom *Lentinus edodes*, is a β-(1→3), β-(1→6)-D-glucan with effective antitumor and immunopotentiating activity (Vannucci et al. 2013). Molecular weight of lentinan is particularly large, on the order of 400,000- 1,000,000 daltons. Its primary structure is a (1→3)-β-glucan

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consisting of five (1→3)-β-glucose residues in a linear linkage and two (1→6)-β-glucopyranoside branches in side chains which result in a right-handed triple helical structure (Chihara, 1992).

Lentinan, an antitumour drug inhibits the tumour growth by stimulating the immune system but is quite nontoxic to tumour cells (Chihara et al., 1970; Chihara, 1978). The stimulation of immune system occurs when β-D-glucan binds to lymphocyte surfaces or serum specific proteins which activate the macrophages, T-helper cells, NK and other effector cells. The activation of the effector cells results in an increased amount of antibodies production as well as interleukins (IL-1, IL-2) and interferon (IFN-¥) (Mizuno, 1995a; Mizuno, 1995b). It acts as the host defense potentiator and is able to store or augment the responsiveness of host cells by stimulating maturation, differentiation or proliferation of cells involved in host defense mechanism (Chihara et al., 1970). Prominent antitumor activity is exhibited against allogeneic, syngeneic and autochthonous tumors in mice (Zakany et al., 1980; Suga et al., 1984). Lentinan showed prominent antitumour activity not only against allogenic tumours, such as sarcoma 180, but also against various synergic and autochthonous tumours. It prevents chemical and viral oncogenesis and also suppresses cancer metastasis and recurrence in animal models (Zakany et al., 1980; Suga et al., 1984). Additionally, Lentinan is effective against various semisyngeneic tumors, such as P815 mastocytoma, L-5178Y lymphoma, virus induced MM-46, and MM-102 carcinoma (Chihara, 1992; Ochiai et al., 1992). Lentinan has been fruitful in extending the overall survival rate of cancer patients, particularly those having gastric and colorectal carcinomas (Furue et al., 1981; Taguchi et al., 1985a; 1985b). Studies undertaken by Chihara et al (1969) suggested that lentinan is not only able to raise the host resistance against different types of cancer but also has the ability to revive the immune function of affected individuals. Lentinan enhances the antitumor cytotoxic activity of peritoneal macrophages against human melanoma target cells *in vitro* (Ladanyi et al., 1993). The cumulative activity of (a) lymphokine-activated killer cells stimulated by IL-2, and (b) lentinan against autologous tumour cells and K562 human erythroleukemia cells is greater than that stimulated by IL-2 alone *in vitro*.

Lentinan is also able to restore the suppressed activity of helper T-cells in the tumour-bearing host to their normal state, which leads to complete restoration of humoral immune responses (Maeda et al., 1988). In addition to that, it is reported that the lentinan-induced delayed-type hypersensitivity response are consistent with the antitumor activity of lentinan in the tumour-bearing mice (Suzuki et al., 1994). It is proposed that the delayed-type hypersensitivity response at the tumour sites induced by lentinan and the subsequent infiltration of immune effector cells, such as natural killer cells and cytotoxic T-lymphocytes, into the tumour burden are important mechanism of antitumor action of lentinan. A study shows that lentinan inhibits hepatic metastasis in adenocarcinoma 26-bearing mice by activated Kupffer cells (Taki et al., 1995). Nevertheless, the exact immunomodulatory effect induced by lentinan is a critical aspect that is yet to be explored.

Lentinan has gained the attention for restoring and augmenting responsiveness of host cells while having no direct cytotoxicity against tumours. The antitumor effect of lentinan is lost in the neonatal thymectomized mice and decreased considerably by administration of antilymphocyte serum (Maeda and Chihara, 1971). These results indicate that the antitumor action of lentinan demands an intact T-cell component and this antitumor activity are mediated by a thymus-dependent immune mechanism. Interestingly, the antitumor activity of lentinan is also inhibited by pretreatment with anti-macrophage agents. Thus, the various consequences of lentinan are thought to be

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due to the "potentiation of the response of precursor T-cells and macrophages to cytokines produced by certain classes of lymphocytes, after specific recognition of tumour cells" (Chihara, 1992).

5.4. Protein-bound polysaccharide (PSK)

PSK is a protein bound β-glucan which is produced from the cultured mycelia of *Coriolus versicolor* in Japan (Tsukagoshi et al., 1984; Kobayashi et al., 1995). PSK consists of approximately 62-percent polysaccharide and 38-percent protein, although the percentage of both may differ. It is a (1→4)-β-glucan along with (1→6)-β-glucopyranosidic side chains for every fourth glucose unit. The glucan fragment of PSK comprises of a β-(1→4) main chain and β-(1→3) side chains, with β-(1→6) side chains that is bound to a polypeptide moiety through O or N-glycosidic bonds. It contains acidic amino acids such as aspartic acid and glutamic acid and neutral amino acids such as valine and leucine but basic amino acids such as lysine and arginine are present only in minor quantities. The elemental monosaccharide is glucose with limited amounts of other sugar residues for instance mannose, fucose, xylose and galactose (Tsukagoshi et al., 1984). PSK is a set of molecules whose molecular weight ranges from 94,000 to 100,000 Da as measured by ultracentrifuge analysis.

Clinically PSK is reported to be most beneficial in cancers of the stomach, esophagus, nasopharynx, colon, rectum and lung (Kidd, 2000). It has also shown promise in a subset of breast cancers. Although, PSK is not a patent medicine for cancers, but still it can increase five-year survival in some cases by as much as double and reasonable in some circumstances up to as much as 15 years. PSK can be particularly life-saving in some situations, for instance in case of patients suffering from HLA B40-positive breast cancer or where risk factors such as impaired immunity or high ACT or SA exists (Kikuchi, 1988). Besides, PSK shows a marked antitumor effect against allogeneic tumours such as Sarcoma 180 and Ehrlich carcinoma of experimental animals by both intraperitoneal and oral administration (Tsukagoshi et al., 1984; Kobayashi et al., 1993). The recent study has eventually suggested that PSK exhibits tumoricidal activity by activating the T-cells that identify PSK as an antigen and kills the tumour cells in an antigen specific manner (Okazaki et al., 1995).

Unlike most glucans, PSK (protein-bound polysaccharide) has both direct and indirect cytotoxic effects on tumour cell lines *in vitro*. It was observed that tumour suppression takes place in BDF1 mice when L1210 leukemia cells or P815 mastocytoma cells are mixed with PSK *in vitro* and inoculated subcutaneously (Tsukagoshi et al., 1984). It is reported that PSK enhances the cytotoxic activity of peripheral blood lymphocytes (PBL) against T24 human urinary bladder tumour cell line after PBL are incubated with PSK at concentration of 10-100 mg/ml *in vitro* (Mizutani & Yoshida, 1991; Mizutani et al., 1992). The level of DNA synthesis in PBL increases in the presence of PSK, and the maximum increase is obtained when PBL are cultured with PSK at 100 mg/ml for 5 days (Nio et al., 1991). PSK inhibits growth and DNA synthesis *in vitro* in various cell lines such as L1210 leukemia, P388 leukemia (Yanagawa et al., 1984), Ehrlich carcinoma, Yoshida sarcoma, AH-13 (Tsukagoshi et al., 1984), human hepatoma C-HC-20, human choriocarcinoma GCH-1 and GCH-2 (Kanazawa et al., 1983) and human breast cancer cell MCF-7 (Aoyagi et al., 1997). These results show that antitumor effects of PSK may be, in part, due to its direct cytotoxicity. In ICR mice, PSK administration can revive the antibody production against trinitrophenyl, which has depressed the immunity in Sarcoma 180-bearing mice. In specific-pathogen-free mice, oral dosage of PSK can enhance the

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impaired antitumor $CD4^+$ T-cell response in gut-associated lymphoid tissue (Harada et al., 1997). PSK modulates the cytotoxic activity of peripheral blood lymphocytes *in vivo* and *in vitro*. Local administration of PSK is more efficient than a systemic one (Mizutani & Yoshida, 1991). It is reported that PSK induces gene expression of some cytokines such as TNF-α, IL-1, IL-8 and IL-6, *in vivo* or *in vitro*. PSK also helps to maintain the immune status in spite of the deadly treatment by traditional therapy by stimulating both killer cells *in vivo* and human NK cells in culture, at concentrations reached in the blood by oral administration of 3 grams per day (Mizutani & Yoshida, 1991; Kariya et al., 1992). It has the ability to restore immune potential to the normal level after the host has been depressed by tumour burden or anticancer chemotherapeutic agents (Kobayashi et al., 1993).

5.5. Polysaccharopeptide (PSP)

A PSP has been prepared in China from a strain of *Coriolus versicolor* and also from the ethanol extract of *Trametes versicolor* mycelium (Yang et al., 1992; Yang & Wang, 1994). PSP has not only been confirmed to be non-toxic (Jian et al., 1999; Zhong et al., 1999) but also has remarkable immunopotentiation capacity which is sufficient enough to enhance both the survival rate and quality of life in cancer patients (Jong & Yang, 1999; Yang, 1999). PSP is widely used as an anti-cancer and immunomodulatory agent in clinical medication as this glycopeptide is endowed with antitumour and immunostimulant activities (Yang et al., 1992; Yang & Wang, 1994). PSP has a number of antitumor activities *in vitro*, such as suppressing Ehrlich ascites tumour, leukemia P388, Sarcoma 180 and four different types of human cancer cell lines that is human gastric cancer cells, human lung cancer cells, mononuclear leukemia cells and human skin histiocytic lymphoma cells (Yang et al., 1992). Apart from restricting the enhancement of various human cell lines, it increases the IgG and C3 complement levels in mice and reverses the tumour-induced immunodeficiencies (Yang et al., 1992; Liu et al., 1993). Furthermore PSP demonstrates significant benefits against three cancers – stomach, esophagus, and lung (Jong & Yang, 1999; Yang, 1999). PSP also activates killer cells *in situ* in the living cancer patient. PSP also offers analgesic action, which can be beneficial to the cancer patient (Liu et al., 1999).

Clinical research with PSP has taken a fast track since it was isolated in 1983 (Yang, 1999). PSP has been proven to be non-toxic after completion of Phase I, II, and III human trials (Jong & Yang, 1999; Jian et al., 1999) with marked immunopotentiation capacity sufficient to improve survival rate and quality of life in cancer patients (Jong & Yang, 1999). In the Phase I trial, 16 healthy persons and five breast cancer patients were administered with PSP for one month at doses up to 6 grams per day. According to this trial and other preliminary trials, the appetite was improved in a number of subjects and there was no indication of any negative response (Xu, 1993; Sun et al., 1993). The Phase II and Phase III trials carried out with 58 and 59 subjects respectively proved that PSP is fruitful for patients having stomach, esophageal and non-small-cell lung cancers and simultaneously prevented the undesirable effects caused by the cancers *per se* and the toxic therapies traditionally used to cure them (Sun & Zhu, 1999; Sun et al., 1999). PSP pacified symptoms commonly associated with cancer, including fatigue, anorexia, nausea, thirst, cold sweat and pain. Additionally, PSP also enhanced the severity of systemic toxic deterioration associated with conventional therapies, stabilized or increased body weight, and significantly improved overall immune status.

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5.6. Shizophyllan

Schizophyllan obtained from the culture filtrates of *Schizophyllum commune* is also a (1→3)-β-glucan having a β-glucopyranosyl group linked by 1→6 linkages to every third or fourth residue of the main chain. It is similar to lentinan in its triple helix structure and biological activity, but physicochemically dissimilar to lentinan. Its molecular weight is about $450x10^3$ Da (Komatsu et al., 1969; Ohno et al., 1995). Schizophyllan inhibits solid Sarcoma 180 tumour when injected by intraperitoneal or intravenous route, but has low antitumor activity by subcutaneous route (Komatsu et al., 1969). The shizophyllan derived from *Shizophyllan commune*, shown to activate macrophages *in vitro* and *in vivo*, is responsible for augmentation of T-cell activities and increases sensitivity of cytotoxic LAK and NK cells to IL-2 (Mizuno et al., 1990). Further, the induction of gene expression of cytokines by schizophyllan has been studied *in vitro* and *in vivo* (Nemoto et al., 1993; Okazaki et al., 1995). After schizophyllan is administered intraperitoneally in ICR mice, the kinetics of gene expression of cytokines is different in peritoneal exudate cells, splenocytes and liver cells. It is generally accepted that protein synthesis and gene expression of cytokines are regulated separately. Therefore, the antitumor activity of schizophyllan is mainly due to host-mediated immune responses (Nemoto et al., 1993; Okazaki et al., 1995).

Schizophyllan has prolonged the survival time in patients having head and neck cancer but it was found to be comparatively ineffective in gastric cancer (Kimura et al., 1994; Borchers et al., 1999). In a randomized controlled study of schizophyllan in combination with radiotherapy, schizophyllan significantly prolonged the overall survival of stage II cervical cancer patients but not stage III (Okamura et al., 1986; 1989). In a prospective, randomized clinical trial involving 312 patients treated with surgery, radiotherapy, chemotherapy (fluorouracil) and schizophyllan in various combinations, patients treated with schizophyllan had a better overall survival than patients who had not received the polysaccharides (Miyazaki et al., 1995).

6. Summary and future prospects

In the last few decades, large numbers of mushroom have been increasingly used as a source of medicinal compounds and therapeutic adjuvants or health food supplements. Recently, the biological activities of polysaccharides or polysaccharide-protein complexes derived from mushrooms have received much attention as prebiotics and in biomedical sciences. Functional compounds like prebiotics from mushrooms in the diet seem to be an attractive alternative to increase life quality with low the risk of obesity, cancer, hypersensitivity, vascular diseases and degenerative ailments. Besides these, the most promising activities of prebiotic polysaccharides from mushrooms are immunomodulation and anti-cancer effects. However, the mode of action for antitumor nature of polysaccharides or polysaccharide-protein complexes isolated from mushrooms is still not clearly understood. It is widely accepted that antitumor polysaccharides from mushrooms enhance various immune responses *in vivo* and *in vitro*, and act as biological response modifiers. Thus extensive studies need to be conducted to develop food supplements from mushrooms for imparting prebiotic effects. Along with benefiting from the prebiotic effect of mushroom extracts, consumers will also be able to obtain medicinal advantages from these valuable compounds.

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Table 1 - Compounds and their biological properties.

Compounds	Biological source	Biological activity	References
(1→3)-,(1→6)-branched glucan, a heteroglycan 9 (Fr. II) consisting of D-mannose, D-glucose, and D-galactose, a (1→6)-a-glucan and a (1→3)-, (1→6)-β-D-glucan	*Pleurotus florida*	Immunoenhancing, stimulates macrophages, splenocytes and thymocytes	Roy et al., 2009
2 polysaccharide fractions (IA-a and IA-b). IA-a is a 1,4-α-glucan exhibiting 1,6-branching.This branching structure suggests that IA-a has a glycogen-like structure. The IA-b component appears to be an arabinoxylan-like polysaccharide, mainly consisting of arabinose and xylose.	*Lentinula edodes*	Potently stimulated cytokine production, stimulate phagocytosis	Kojima et al., 2010
Branched (1→3)-β-D-glucan	*Dictyophora indusiata*	Anti-inflammatory effects	Hara et al., 1982
D-glucopyranose	*Lentinus edodes*	Immunomodulating activities	Zheng et al., 2005
Glucan	*Pleurotus florida*	Immunomodulating	Rout et al., 2004
Glucan	Somatic hybrid mushroom PCH9FB of *Pleurotus florida* and *Calocybe indica var.*	Immunoenhancing and antioxidant properties with immune activation of macrophage, splenocyte, and thymocyte	Maity et al., 2011
Glucan	*Lentinus squarrosulus*	Optimum activation of macrophages as well as splenocytes and thymocytes	Bhunia et al.,2011
Glucan	*Calocybe indica*	Stimulates the splenocytes and thymocytes	Mandal et al.,2012
Glucan	*Tricholoma crassum*	Macrophage activation in vitro by NO production in a dose dependent manner and strong splenocyte and	Samanta et al., 2012

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		thymocyte immunostimulation in mouse cell culture Medium; also exhibited good inhibition activity toward lipid peroxidation.	
Glucan consisting of terminal, (1→3,6)-linked, and (1→6)-linked β-D-glucopyranosyl moieties	Hybrid mushroom Pfle1r of *Pleurotus florida* and *Lentinula edodes*	Macrophages, splenocytes, and thymocytes activation	Maji et al.,2012
Glucans	*Russula albonigra* (Krombh.)	Excellent activations of macrophages as well as splenocytes and thymocytes *in vitro*.	Nandi et al., 2012
Glucans (PS-I, PS-II and PS-III)	*Pleurotus florida*	Antitumor and immunomodulating properties	Ojha et al., 2010
Glucomannan	*Agaricus blazei*	Antitumour activity	Mizuno et al.,1999
Hetero polysaccharide	*Tricholoma crassum*	Splenocyte, thymocyte as well as macrophage activations	Patra et al., 2012
Heteroglucans, glycoprotein	*Agaricus blazei*	Remarkable antitumour activity	Mizuno et al.,1990
Heteroglycan	*Lentinus squarrosulus*	Immunoenhancing	Bhunia et al., 2010
Heteroglycan	Somatic hybrid mushroom (PfloVv1aFB), raised through protoplast fusion between the strains of *Pleurotus florida* and *Volvariella volvacea*	Immunoenhancing properties, macrophage, splenocyte, and thymocyte activation	Patra et al., 2011
Heteroglycan consisting of (1→3)-, (1→6)-, (1→3,4)-linked, and terminal β-D-Glcp along with (1➞2,6)-α-D-Galp and terminal α-D-Manp	Somatic hybrid mushroom (PfloVv1aFB), raised through protoplast fusion between the strains of *Pleorutus florida* and *Volverilla*	Strong immunostimulating activity of macrophages as well as splenocytes and thymocytes	Bhunia et al., 2012

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	volvacea		
Heteropolysaccharide	*Pleurotus ostreatus*	Stimulates macrophages, splenocytes, and thymocytes.	Maity et al., 2011
Lentinan, LC-11, LC-12, LC-13, EC-11 and EC-14	*Lentinus edodes*	Antitumor effect; lentinan especially markedly inhibited the growth of Sarcoma 180 implanted s.c. in mice.	Chihara et al.,1970
Novel polysaccharide consisting of galactose, fucose, and glucose	Somatic hybrid mushroom of *Pleurotus florida* and *Calocybe indica* variety APK2	Macrophage, splenocyte, thymocyte activation as well as antioxidant property	Maity et al., 2011
Polysaccharide	Grifola frondosa (Maitake)	Antitumour activity	Nanba et al.,1987
Polysaccharide consisting of D-glucose and D-galactose	*Pleurotus florida*	Immunoenhancing	Dey et al., 2010
Polysaccharide LT1 which has a backbone chain composed of 1→4-linked and 1→3-linked glucopyranosyl residues and has branches of single glucosyl stubs at C-6 of β-(1→4)-linked glucopyranosyl.	*Lentinula edodes*	Natural antitumor drug	Yu et al., 2010
Two glucans (PS-I and PS-II) isolated	*Pleurotus florida*	PS-I showed macrophage, splenocyte and thymocyte activations	Dey et al., 2012
Two glucans (watersoluble PS-I, water-insoluble PS-II) were isolated. PS-I was found to consist of only (1→6)-linked β-D-glucopyranose. PS-II was composed of terminal, (1→3,4)-linked, and (1→3)-	Hybrid mushroom (backcross mating between PfloVv12 and *Volvariella volvacea*)	PS-I showed macrophages, splenocytes, and thymocytes activation as well as antioxidant property	Sarkar et al., 2012

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linked β-D-glucopyranosyl moieties.			
Varying amounts of both α- and β-glucans	*Agaricus bisporus, Agaricus brasiliensis, Phellinus linteus and Ganoderma lucidum*	Antioxidative and immunomodulating activities	Kozarski et al., 2011
α-1,6- and α-1,4-glucan	*Agaricus blazei*	Stimulate lymphocyte T-cell subsets in mice	Mizuno et al.,1998
β-D-glucan	*Lentinus edodes*	Antitumour effect	Sasaki and Takasuka, 1976
β-D-glucans	*Collybia dryophila*	Strongly inhibit nitric oxide production in activated macrophages suggesting that this polysaccharide displays a potential anti-inflammatory activity, potent bioactive fungal compounds	Pacheco-Sanchez et al., 2006
β-glucan	Somatic hybrid (PfloVv5FB) of *Pleurotus florida* and *Volvariella volvacea*	Strong immunoactivation of macrophages, splenocytes as well as Thymocytes.	Maity et al., 2013
β-glucans	*Termitomyces robustus*	Immunostimulating, significant macrophage, splenocyte, and thymocyte activation	Bhanja et al., 2012

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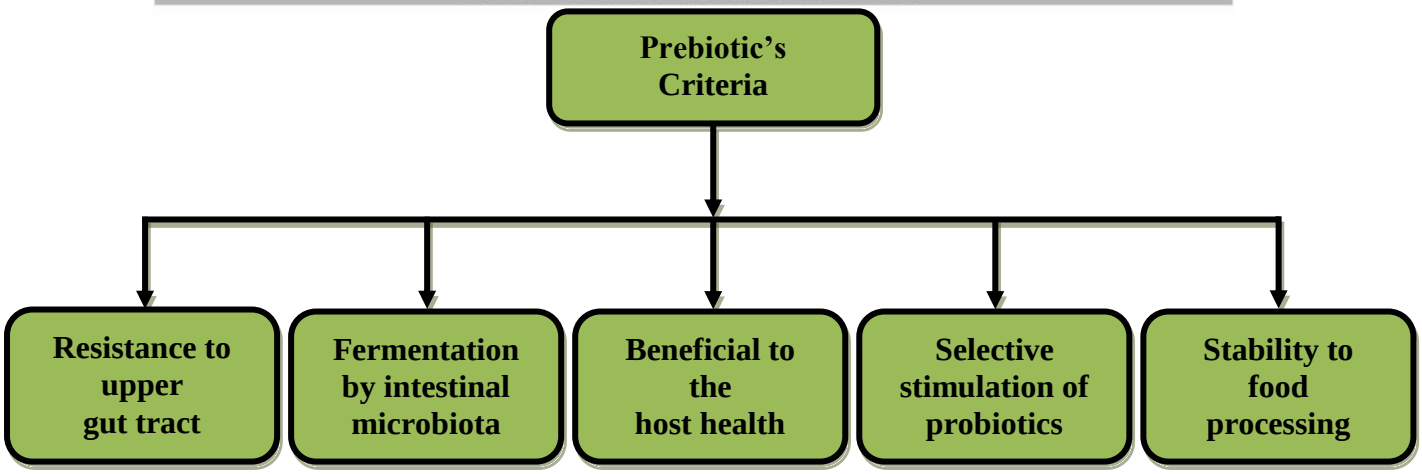

Fig. 1 - Criteria for prebiotic compounds (Wang, 2009).

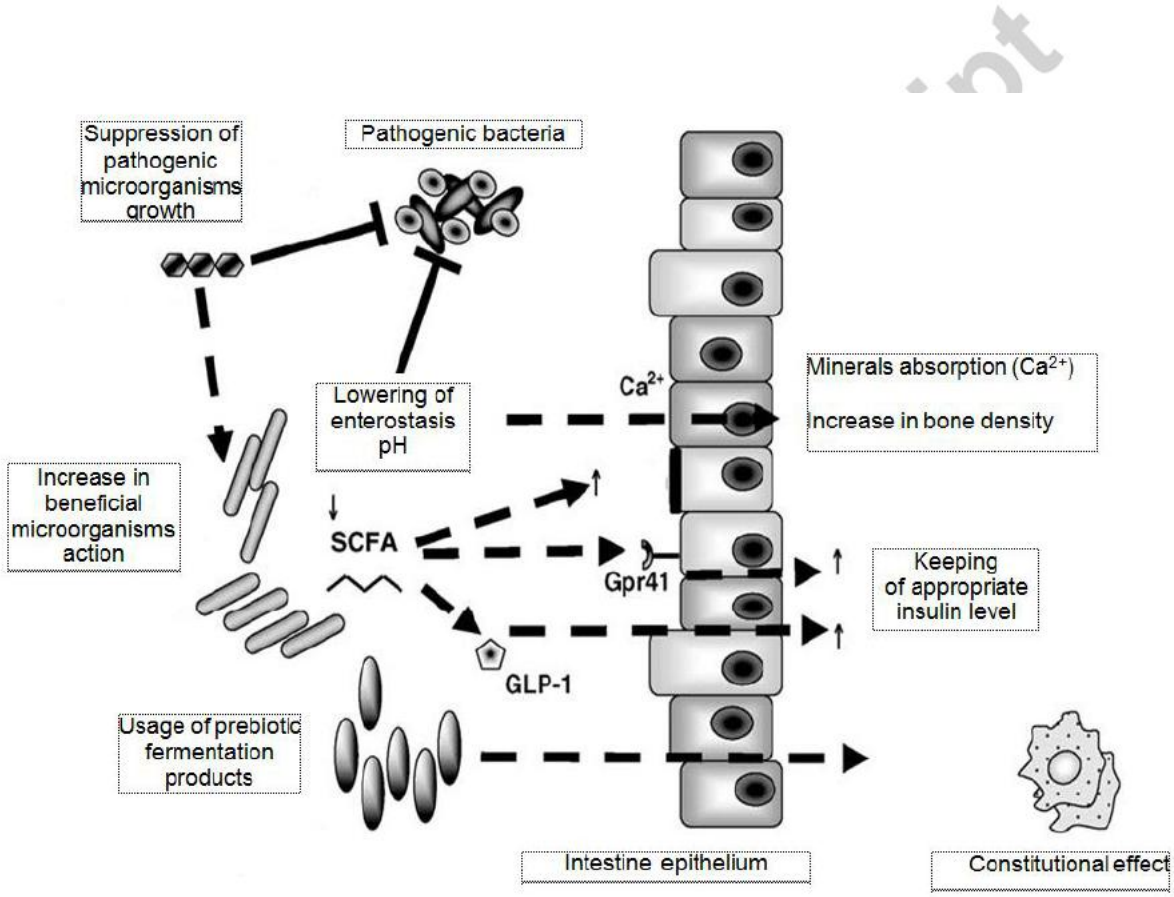

Fig. 2 - Mechanism action of prebiotics (Śliz□ewska et al., 2012).

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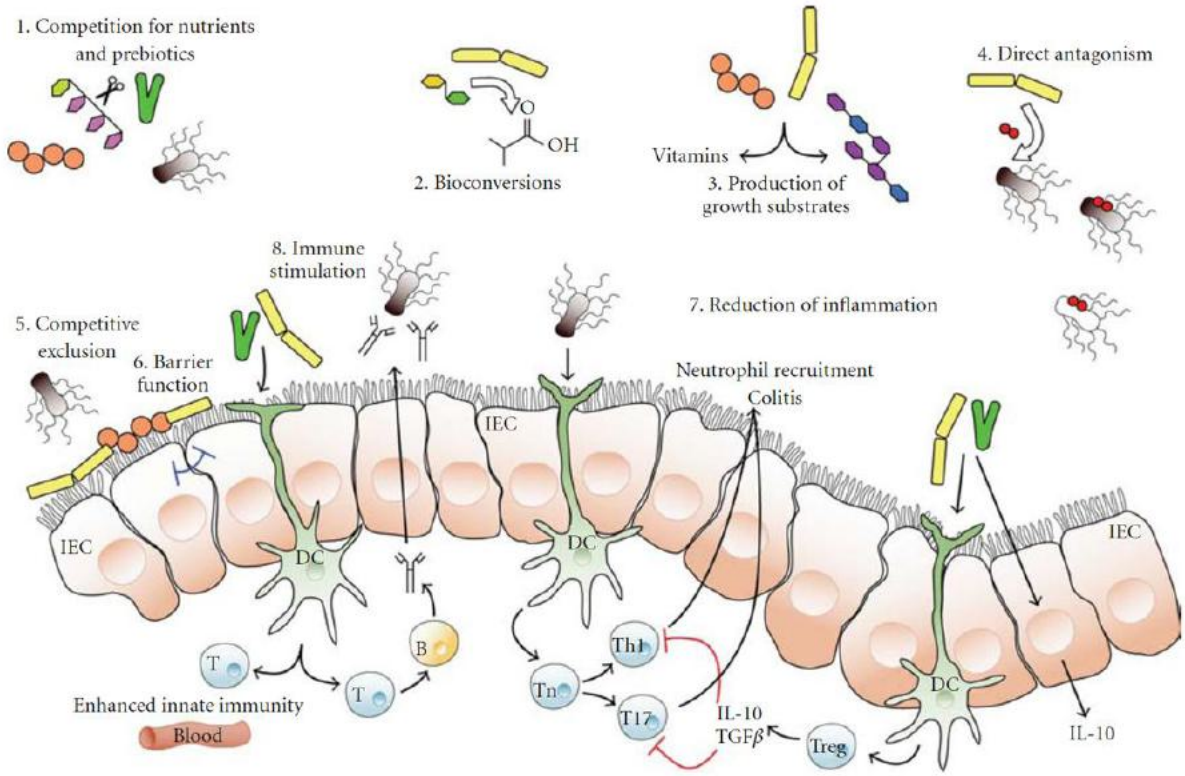

Fig. 3 - Potential or known mechanisms of probiotics action (O'Toole & Cooney, 2008). These mechanisms include (1) competition for dietary ingredients as growth substrates, (2) bioconversion of, for example, sugars into fermentation products with inhibitory properties, (3) production of growth substrates, for example, EPS or vitamins, for other bacteria, (4) direct antagonism by bacteriocins, (5) competitive exclusion for binding sites, (6) improved barrier function, (7) reduction of inflammation, thus altering intestinal properties for colonisation and persistence within, and (8) stimulation of innate immune response (by unknown mechanisms). IEC, epithelial cells; DC, dendritic cells; T, T cells.

→3)-α-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→
6
↑
1
α-D-Glc*p*

Fig. 4 - Repeating unit of glucan isolated from *Pleurotus florida* (Roy et al., 2009).

→3)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→
6
↑
1
β-D-Glc*p*

Fig. 5 - Repeating unit of glucan isolated from *Lentinus squarrosulus* (Bhunia et al., 2011).

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→[6)-β-D-Glc*p*-(1]$_2$→4)-β-D-Glc*p*-(1→[4)-α-D-Glc*p*-(1]$_2$→
6
↑
1
β-D-Glc*p*

Fig. 6 - Repeating unit of glucan isolated from *Calocybe indica* (Mandal et al., 2012).

→3)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→3)-α-D-Glc*p*-(1→
6
↑
1
α-D-Glc*p*-(3→1)-α-D-Gls*p*

Fig. 7 - Repeating unit of glucan isolated from *Russula albonigra* (Nandi et al., 2012).

→6)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→
3
↑
1
β-D-Glc*p*-(1→3)-β-D-Gls*p*

Fig. 8 - Repeating unit of glucan isolated from *Ramaria botrytis* (Bhanja et al., 2013).

→3)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→
6
↑
1
β-D-Gls*p*

Fig. 9 - Repeating unit of glucan isolated from *Tricholoma crassum* (Samanta et al., 2013).

F **D** **A** **C** **E**
→6)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→6)-α-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→4)-β-D-Glc*p*-(1→
4 6
↑ ↑
α-L-Fuc*p* β-D-Glc*p*
B **G**

Fig. 10 - Repeating unit of heteroglycan isolated from *Lentinus squarrosulus* (Bhunia et al., 2010).

→3)-α-D-Gal*p*-(1→4)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→
6
↑
1
α-L-Fuc*p*

Fig. 11 - Repeating unit of heteroglycan isolated from *Calocybe indica* (Mandal et al., 2011).

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C **E** **D** **C** **E** **D**
→3)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→6)-β-D-Glc*p*-(1→
3 3
↑ ↑
α-D-Gal*p* α-D-Glc*p*
A **B**

Fig. 12 - Repeating unit of heteroglycan isolated from *Pleurotus ostreatus* (Maity et al., 2011).

→6)-β-D-Glc*p*-(1→2)-α-D-Gal*p*-(1→3)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→
6 6
↑ ↑
1 1
α-D-Man*p* β-D-Glc*p*

Fig. 13 - Repeating unit of heteroglycan isolated from hybrid mushroom (*PfloVv1aFB*) (Patra et al., 2011).

→6)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→2)-α-D-Gal*p*-(1→3)-β-D-Glc*p*-(1→
4 6
↑ ↑
1 1
β-D-Glc*p* α-D-Man*p*

Fig. 14 - Repeating unit of heteroglycan isolated from hybrid mushroom (*PfloVv1aFB*) (Bhunia et al., 2012).

→6)-α-D-Gal*p*-(1→6)-α-D-Glc*p*-(1→
2
↑
1
β-D-Man*p*

Fig. 15 - Repeating unit of heteroglycan isolated from hybrid mushroom (*Pfle1r*) (Maji et al., 2013).

C
β-D-Man*p*
1
↓
B 2 **A**
→6)-α-D-Gal*p*-(1→6)-α-D-Gal*p*-(1→
4
↑
1
β-D-Glc*p*
D

Fig. 16 - Repeating unit of heteroglycan isolated from hybrid mushroom (*Pfle1r*) (Sen et al., 2013).

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D **B** **A**

→6)-β-D-Glc*p*-(1→6)-α-D-Glc*p*-(1→6)-α-D-Glc*p*-(1→

3

↑

1

α-D-Gal*p*-(4←1)-β-D-Man*p*

C **E**

Fig. 17 - Repeating unit of hetero-polysaccharide isolated from *Tricholoma crassum* (Patra et al., 2012).

→3)-β-D-Glc*p*-(1→3)-β-D-Glc*p*-(1→6)-α-D-Glc*p*-(1→6)-α-D-Gal*p*-(1→6)-α-D-Gal*p*-(1→

4 2

↑ ↑

1 1

β-D-Glc*p* β-D-Man*p*

Fig. 18 - Repeating unit of hetero-polysaccharide isolated from hybrid mushroom (*pfle 1p*) (Maity et al., 2013).

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